Formula transcription fixture

The formulas below are separated by ordinary prose so their reading regions can be identified.

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y \tag{1}$$

The residual sum of squares has indexed terms and limits on a displayed sum.

$$\mathrm{RSS}(\beta) = \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 \tag{2}$$

The derivative below contains a fraction rule and a superscript inside an operator's argument.

$$\frac{\partial}{\partial x} \sin(x^2) = 2x \cos(x^2) \tag{3}$$

The entries of a two by two matrix must retain their row and column positions.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{4}$$

The next expression places limits above and below an integral.

$$\int_0^{\infty} e^{-x}\, dx = 1 \tag{5}$$

The radical's rule covers both terms of its argument.

$$r = \sqrt{x^2 + y^2} \tag{6}$$

The final expression contains a fraction nested inside another fraction.

$$f(x) = \frac{1}{1 + \frac{1}{x}} \tag{7}$$